

Oral Exam Preparation Sheet

Fall 2025

1. Basics of Relational Databases (RDBMS)

- Subtopics:
 - Tables, Rows, Columns
 - Primary Keys, Foreign Keys
 - Normalization vs. Denormalization
- Sample Questions:
 - What is the purpose of a primary key?
 - Explain the concept of referential integrity.
 - What are the advantages of normalization?

2. Structured Query Language (SQL)

- Subtopics:
 - SELECT, INSERT, UPDATE, DELETE
 - JOIN types
 - GROUP BY and Aggregate Functions
- Sample Questions:
 - Write a query to find the top 5 highest paid employees.
 - What is the difference between INNER JOIN and LEFT JOIN?
 - How do you use GROUP BY and HAVING together?

3. NoSQL Databases

- Subtopics:
 - Key-Value Stores, Document Stores
 - Columnar Databases
 - Graph Databases
- Sample Questions:
 - What are the main categories of NoSQL databases?
 - When would you prefer a NoSQL over an RDBMS?
 - Name a real-world use case for a graph database.

4. MongoDB Overview

- Subtopics:
 - Collections and Documents
 - JSON/BSON format
 - CRUD Operations in MongoDB
- Sample Questions:
 - How does MongoDB store data?
 - What are collections in MongoDB?
 - Write a basic MongoDB query to find all documents with age > 30.

5. Clustering and Partitioning in MySQL

- Subtopics:
 - Clustered Index
 - Partition Types (RANGE, LIST, HASH)
 - Query Optimization
- Sample Questions:
 - What is a clustered index and how does it improve performance?
 - Explain the difference between partitioning and clustering.
 - How does MySQL handle range queries with clustered indexes?

6. Data Lake vs. Data Warehouse

- Subtopics:
 - Structured vs. Semi-Structured Data
 - ETL vs. ELT
 - Storage and Performance Differences
- Sample Questions:
 - What is the key difference between a Data Lake and a Data Warehouse?
 - Which one is better for raw IoT data storage and why?
 - Give an example use case for each.

7. Query Optimization Techniques

- Subtopics:
 - EXPLAIN Plan

- Indexes
- Avoiding SELECT *
- Limiting rows
- Sample Questions:
 - How does using an index improve performance?
 - What does EXPLAIN do in MySQL?
 - Why should you avoid SELECT * in queries?

8. Graph Databases (e.g., Neo4j)

- Subtopics:
 - Nodes and Edges
 - Use cases (social networks, fraud detection)
 - Cypher Query Language
- Sample Questions:
 - What are graph databases used for?
 - How does Cypher compare to SQL?
 - Give an example of a query to find all friends of a user.

9. CAP Theorem in Databases

- Subtopics:
 - Consistency, Availability, Partition Tolerance
 - Trade-offs in RDBMS vs NoSQL
 - MongoDB and CAP
- Sample Questions:
 - Explain the CAP theorem with examples.
 - Which CAP properties does MongoDB favor?
 - How does network partitioning affect database guarantees?